

THE PARTIAL RESULT SEPARATES A STRONGER AND A WEAKER REGIME

Official BigCodeBench 0.1.5 evaluation; $\alpha=3$; continuous last-token steering

DEEPSEEK

Features evaluated 28/35
Baseline 0/80
Feature-task transitions 93
Unique corrected tasks 14/80

All results use the canonical per-task seed rule.

CODELLAMA

Features evaluated 32/32
Baseline 0/50
Feature-task transitions 6
Unique corrected tasks 4/50

All results use the canonical per-task seed rule.

PROMISING

DeepSeek contains several multi-task causal candidates.

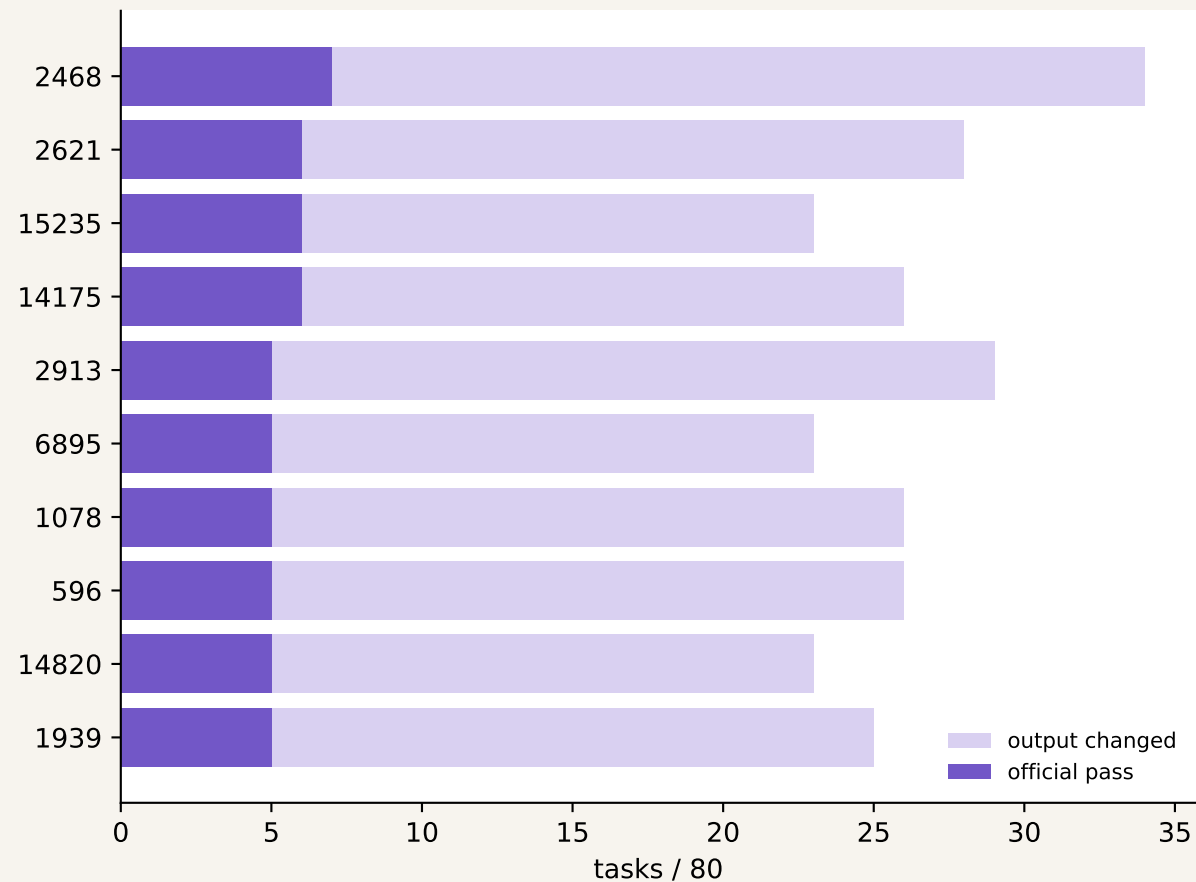
WEAKER / DIFFUSE

CodeLlama yields sparse, task-specific corrections.

PROVISIONAL · DeepSeek is paused at 28/35; CodeLlama is complete at 32/32.

DEEPSEEK: THE BEST FEATURES PRODUCE MULTIPLE OFFICIAL CORREC

Causal outcome first; screening E/V breaks ties



TOP 5

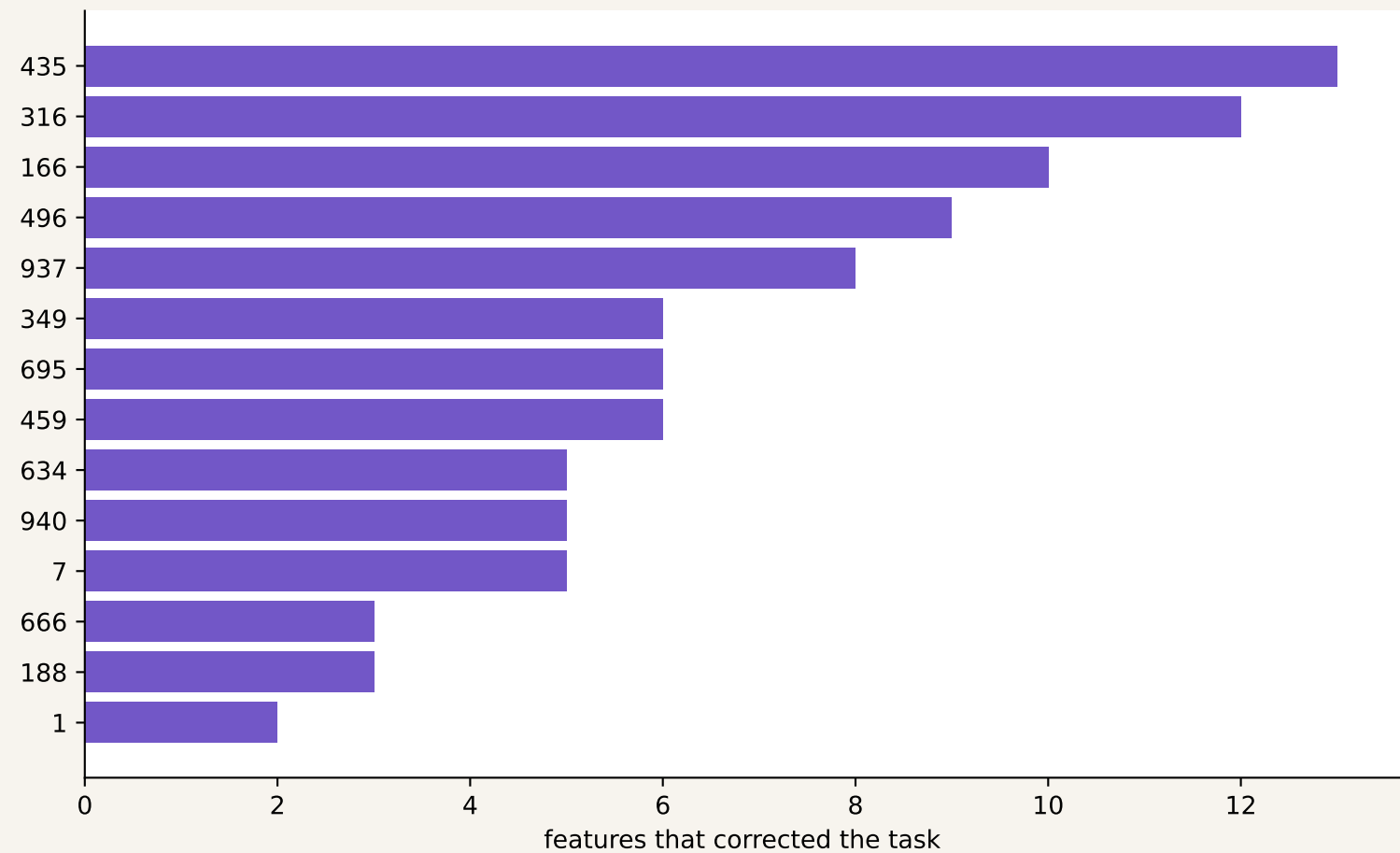
1. 2468 7/80 pass 34/80 changed E/V -7.29
2. 2621 6/80 pass 28/80 changed E/V -10.53
3. 15235 6/80 pass 23/80 changed E/V -4.61
4. 14175 6/80 pass 26/80 changed E/V +4.09
5. 2913 5/80 pass 29/80 changed E/V -10.95

Changing many outputs is not sufficient:

14295 changed 35/80 but produced only 2 official passes.

DEEPSEEK: 93 TRANSITIONS COLLAPSE TO 14 SUSCEPTIBLE TASKS

Feature-task outcomes are not independent causal successes



CONCENTRATION

/435

13 features

/316

12 features

/166

10 features

/496

9 features

INTERPRETATION

The screen finds causal levers, but some tasks are broadly steering-susceptible

FEATURE 2468 IS THE LEADING DEEPSEEK CANDIDATE

Meaning ↔ failure mode ↔ activation timing must remain distinct

SCREENING

paired local · rank 1
E/V -7.29
summary: max
selected around the late divergence window

CAUSAL OUTCOME

7/80 official passes
34/80 outputs changed
13/80 contamination cleanups
baseline: 0/80

CAUSED TIMING

Among changed outputs:
mean first divergence: 42.7%
median: 40.8%

Continuous steering can redirect earlier than the natural activation window

SUPPORTED local screening nominated a feature with repeated official causal effects.

NOT YET SUPPORTED feature-specificity versus sham/matched latent controls.

CODELLAMA: SINGLE-FEATURE CONTROL IS WEAK AND TASK-SPECIFIC

Complete 32-feature sweep; baseline 0/50

SIX FEATURES · ONE PASS EACH

7692 → BigCodeBench/119
10818 → BigCodeBench/119
5642 → BigCodeBench/630
11596 → BigCodeBench/119
4309 → BigCodeBench/490
8142 → BigCodeBench/823

FOUR UNIQUE TASKS

/119 corrected by 3 features
/823 corrected by 1
/490 corrected by 1
/630 corrected by 1

26/32 features produced no official correction.

Trajectory changes occur earlier and more diffusely than in the contamination-focused DeepSeek cohort.

Boundary condition: correlation and trajectory control do not imply that one latent can repair a heterogeneous merged-model regression.

NEXT TARGETS COMBINE CAUSAL EFFECT WITH SCREENING STRENGTH

Ordering: official passes first; |E/V| breaks ties

DEEPSEEK

RANK	FEATURE	PASS	BEST SCREEN
1	2468	7/80	paired local · #1
2	2621	6/80	paired global · #8
3	15235	6/80	association global · #8
4	14175	6/80	association local · #1
5	2913	5/80	paired global · #1

CODELLAMA

E/V	RANK	FEATURE	PASS	BEST SCREEN	E/V
7.29	1	7692	1/50	association global · #1	5.90
10.53	2	10818	1/50	association global · #2	5.90
4.61	3	5642	1/50	association global · #3	5.74
4.09	4	11596	1/50	association global · #5	5.61
10.95	5	4309	1/50	association local · #9	4.86

These are candidates for full dose curves plus sham and randomly selected/matched latent controls—not confirmatory winners yet.